Experiment – 2

Aim :

Implementation and Time analysis of linear and binary search algorithm.

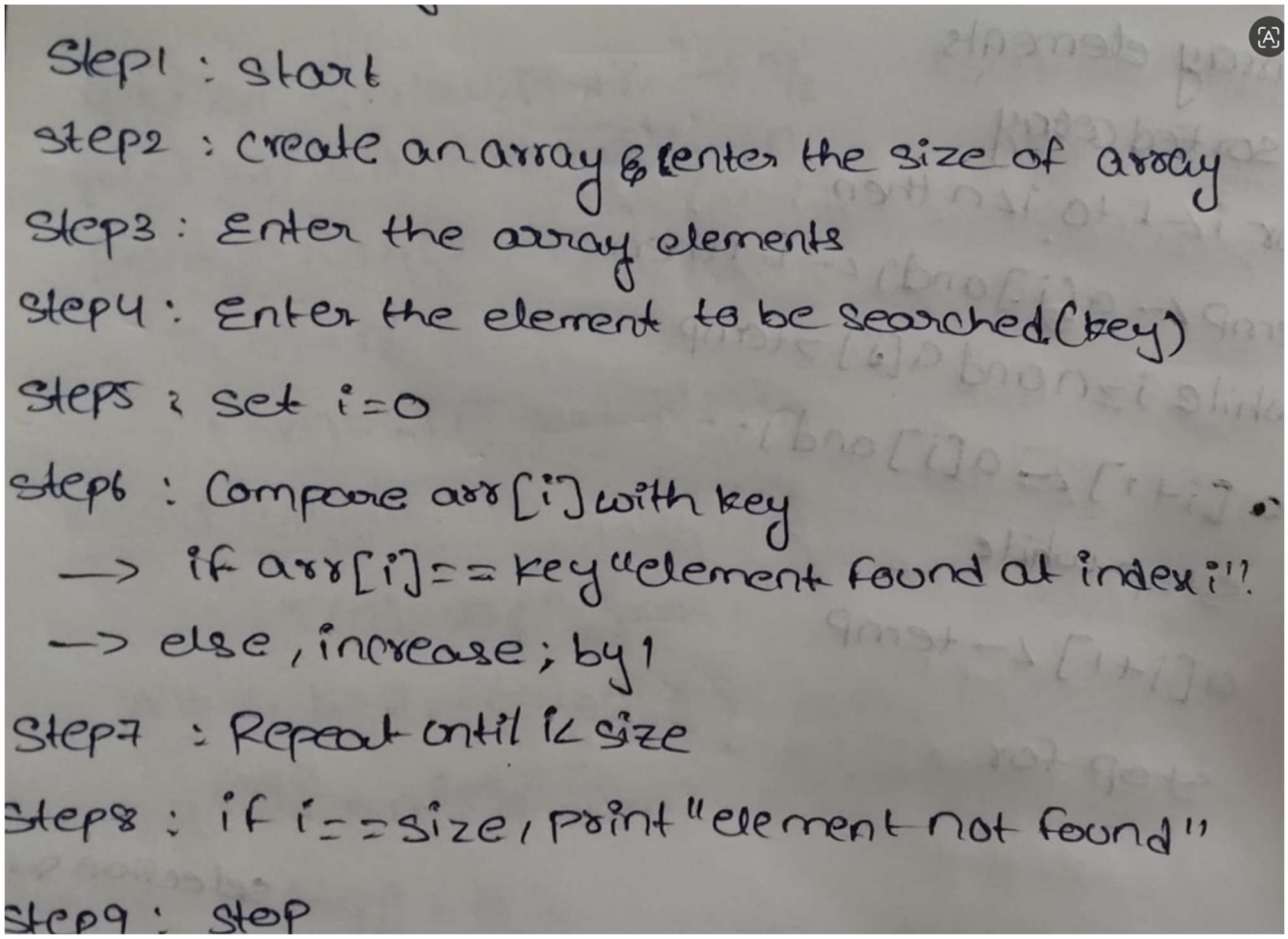
Hardware Required:

Mouse, keyboard

Software Required:

code Blocks

a) Linear search Algorithm:



Program:

```
#include<iostream> using
namespace std;

class linearsearch {
public: int arr[100], size,
key;

   void arrayinput() {
      cout << "enter size of array: ";
      cin >> size; cout << "enter
      array elements: "; int i; for(i =
      0; i < size; i++) {
         cin >> arr[i];
      }
   }

   void printarray() { int i;
      for(i = 0; i < size; i++)
      {
         cout << arr[i] << " ";
      } cout <<
   endl; }

   void searchlogic() { int i;
      for(i = 0; i < size; i++)
      {
         if(arr[i] == key) {
            cout << "element found at index " << i << endl;
            break;
         }     }
      if(i    ==
      size)    {
      cout  <<
```

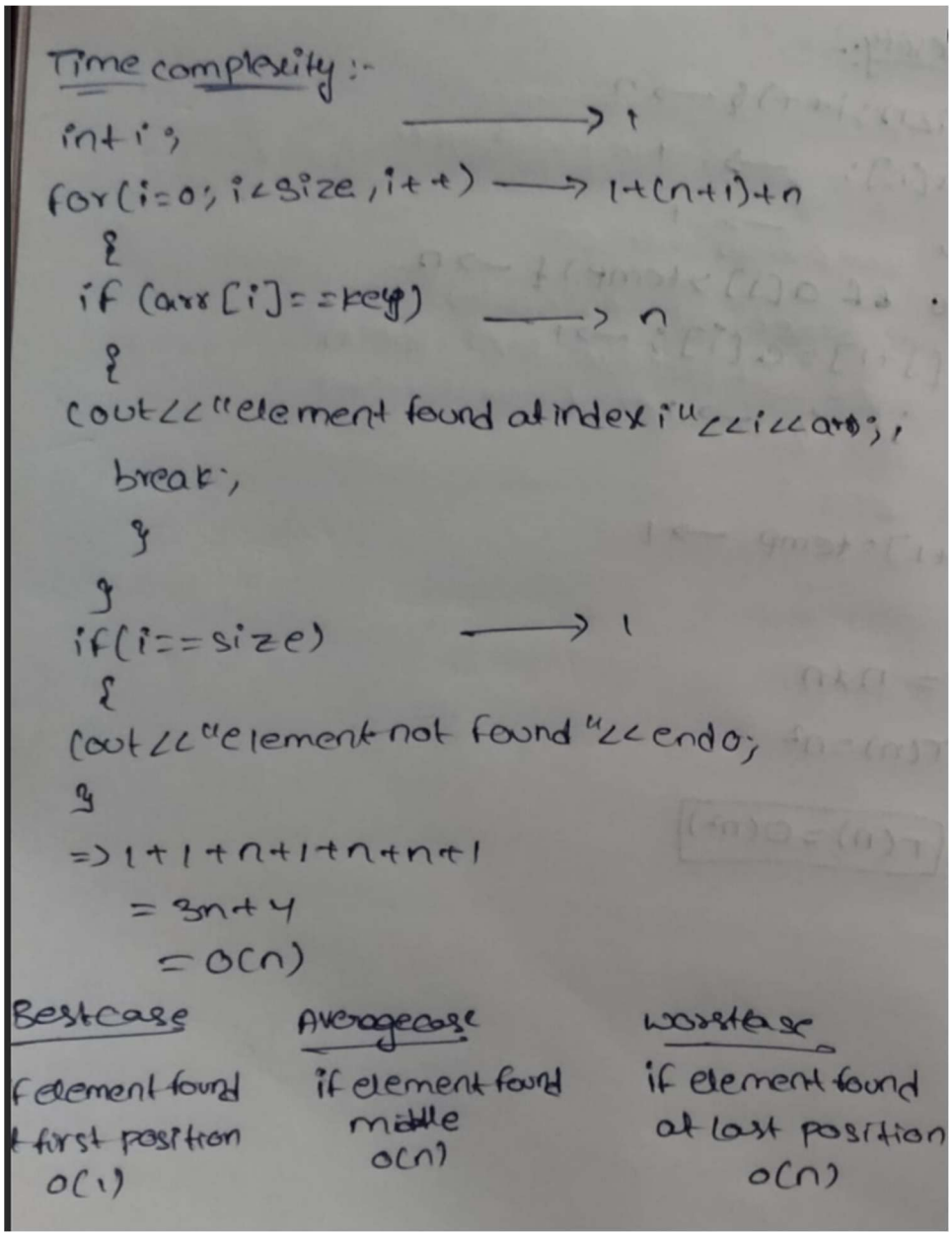
```
        "elemen
        t       not
        found"
        << endl;
        }
    } };

int main() { linearsearch
    ls; ls.arrayinput();
    cout << "array elements" << endl;
    ls.printarray();
    cout << "enter element to search: ";
    cin >> ls.key;
    ls.searchlogic();
    return 0;
}
```

Output:

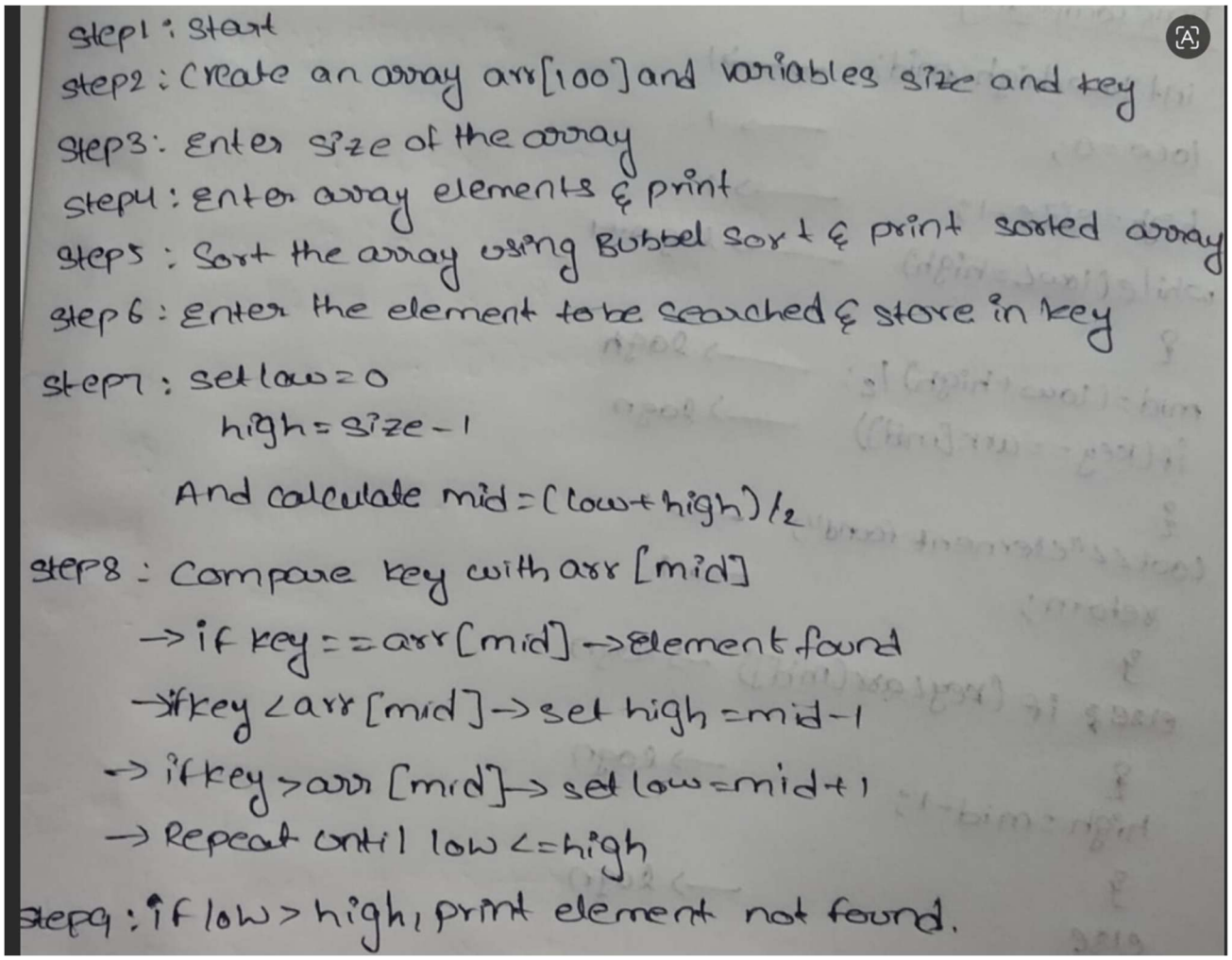
```
enter size of array: 6
enter array elements: 5
2
4
7
9
8
array elements
5 2 4 7 9 8
enter element to search: 4
element found at index 2
```

Time complexity:



b) Binary Search

Algorithm:



Program:

```
#include<iostream> using
namespace std;

class binarysearch {

public: int

arr[100],size,key;

void arrayinput()
{ cout<<"enter size of array: ";

   cin>>size; cout<<"enter array

   elements: ";
```

```
    int i;
    for(i=0;i<size;i++)
    { cin>>arr[i];
    } }

void printarray()
{ int i;
    for(i=0;i<size;i++)
    { cout<<arr[i]<<" ";
} cout<<endl; }

void bubblelogic()
{ int i,j,temp;
  for(i=0;i<size-1;i++)
    { for(j=0;j<size-i-1;j++)
        { if(arr[j]>arr[j+1])
            { temp=arr[j];
              arr[j]=arr[j+1];
              arr[j+1]=temp;
            }
        }
    } }

void searchlogic()
{ int low,high,mid;

    low=0; high=size-
    1;

    while(low<=high)
    { mid=(low+high)/2;

        if(key==arr[mid])
        { cout<<"element found";
        return; } else
        if(key<arr[mid])
        { high=mid-
        1; } else {
        low=mid+1;
        }
```

```
    } cout<<"element not
    found";
}
}; int

main() {

binarysea

rch bs;

bs.arrayi

nput();
bs.bubble

logic();


    cout<<"sorted array"<<endl;
    bs.printarray();

    cout<<"enter element to search:

    "; cin>>bs.key; bs.searchlogic();

    return 0;
}
```

Output:

```
Enter size of array: 5
Enter array elements: 4
2

3
7
9
Sorted array: 2 3 4 7 9
Enter element to search: 3
Element found at index: 1
```

Time complexity:

```
int low,high,mid;          ---> 1
low=0;                     ---> 1
high=size-1;               ---> 1
while(low<=high)           ---> logn+1
 {
mid=(low+high)/2;          ---> logn
if(key==arr[mid])          ---> logn
 {
cout<<"element found";     ---> 1
 return;                   ---> 1
 }
else if (key<arr[mid])     ---> logn
 {                         ---> logn
 high=mid-1;
 }
 else                      ---> logn
 {
low=mid+1                  ---> logn
 }
}
```

$\Rightarrow 1+1+1+\log n+1+\log n+\log n+1+1+\log n+\log n$

$= 5\log n+5$

$= \log n$

$O(\log n)$

Best case	Average case	Worst case
$O(1)$	$O(\log n)$	$O(\log n)$

Result:

As per the experiment to Implementation and Time analysis of linear and binary search algorithm.

Output was verified successfully according to the given experiment.